Various combinations of these properties can be used to uniquely characterize discrete valuation rings (and hence give alternative definitions).

**Theorem 1.16.** *For an integral domain $A$, the following are equivalent:*

- *$A$ is a DVR.*
- *$A$ is a noetherian valuation ring that is not a field.*
- *$A$ is a local PID that is not a field.*
- *$A$ is an integrally closed noetherian local ring of dimension one.*
- *$A$ is a regular noetherian local ring of dimension one.*
- *$A$ is a noetherian local ring whose maximal ideal is nonzero and principal.*
- *$A$ is a maximal noetherian ring of dimension one.*

*Proof.* See [1, §23] or [2, §9]. □

## 1.6 Integral extensions

Integrality plays a key role in number theory, so it is worth discussing it in more detail.

**Definition 1.17.** Given a ring extension $A \subseteq B$, an element $b \in B$ is *integral over $A$* if is a root of a monic polynomial in $A[x]$. The ring $B$ is *integral over $A$* if all its elements are.

**Proposition 1.18.** *Let $\alpha, \beta \in B$ be integral over $A \subseteq B$. Then $\alpha + \beta$ and $\alpha\beta$ are integral over $A$.*

*Proof.* Let $f \in A[x]$ and $g \in A[y]$ be such that $f(\alpha) = g(\beta) = 0$, where

$$\begin{aligned} f(x) &= a_0 + a_1 x + \cdots + a_{m-1}x^{m-1} + x^m, \\ g(y) &= b_0 + b_1 y + \cdots + b_{n-1}y^{n-1} + y^n. \end{aligned}$$

It suffices to consider the case

$$A = \mathbb{Z}[a_0, \ldots, a_{m-1}, b_0, \ldots, b_{n-1}], \qquad \text{and} \qquad B = \frac{A[x,y]}{\big(f(x), g(y)\big)},$$

with $\alpha$ and $\beta$ equal to the images of $x$ and $y$ in $B$, respectively, since given any $A' \subseteq B'$ we have homomorphisms $A \to A'$ defined by $a_i \to a_i$ and $b_i \to b_i$ and $B \to B'$ defined by $x \mapsto \alpha$ and $y \mapsto \beta$, and if $x + y, xy \in B$ are integral over $A$ then $\alpha + \beta, \alpha\beta \in B'$ must be integral over $A'$.

Let $k$ be the algebraic closure of the fraction field of $A$, and let $\alpha_1, \ldots, \alpha_m$ be the roots of $f$ in $k$ and let $\beta_1, \ldots, \beta_n$ be the roots of $g$ in $k$. The polynomial

$$h(z) = \prod_{i,j}\Big(z - (\alpha_i + \beta_j)\Big)$$

has coefficients that may be expressed as polynomials in the symmetric functions of the $\alpha_i$ and $\beta_j$, equivalently, the coefficients $a_i$ and $b_j$ of $f$ and $g$, respectively. Thus $h \in A[z]$, and $h(x+y) = 0$, so $x+y$ is integral over $A$. Applying the same argument to $h(z) = \prod_{i,j}(z - \alpha_i\beta_j)$ shows that $xy$ is also integral over $A$. □